Survival Analysis

Glossary

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Context

In Survival Analysis (also called *time-to-event*), we model the <u>time until an event occurs</u>.

Key points:

- The event can be death, relapse, disease progression, hospital readmission, machine failure, customer churn, etc.

- The key difficulty is that, in real studies, <u>we often do not observe the event time for everyone</u>:
  - some individuals have not experienced the event by the end of follow-up, or
  - they drop out / are lost to follow-up.

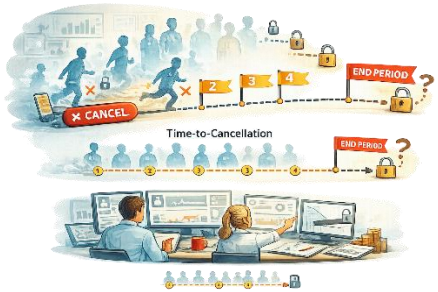
So, the data contains a mix of:

- **exact event times** (when the event is observed), and
- **partial information** of the form: "the event has not happened up to time t."

Examples

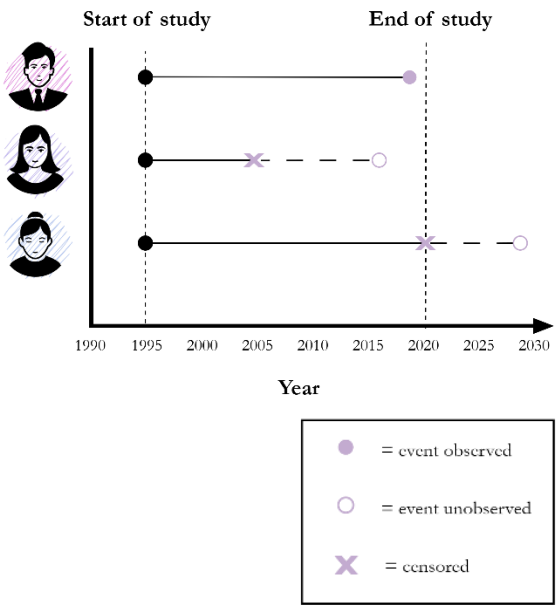
- **Censoring:** For example, suppose that we have conducted a five-year medical study, in which patients have been treated for cancer. We would like to fit a model to predict patient survival time using features such as baseline health measurements and treatment type. At first pass, this may sound like a regression problem. But there is an important complication: hopefully, some or many of the patients have survived until the end of the study. A patient's survival time is said to be censored: we know it is at least 5 years, but we do not know its true value. We do not want to discard this subset of surviving patients, as their survival for at least 5 years is valuable information. However, it is not clear how to make use of this information.

- **Not medical:** For example, consider a company that wishes to model churn, the process by which customers cancel their subscription to a service. The company might collect data over time to model each customer's time-to-cancellation as a function of demographics and other predictors. However, presumably not all customers will have canceled their subscription by the end of this time period; for such customers, the time to cancellation is censored.



- **Not time related:** For instance, suppose we wish to model a person's weight as a function of some covariates, using a dataset with measurements for a large number of people. Unfortunately, the scale used to weigh those people cannot report weights above a certain limit. Then, any weights that exceed that number are censored. The survival analysis methods presented in this chapter could be used to analyze this dataset.

Problem definition



Define two non-negative random variables:

- $\mathrm{T} \geq 0$: the true event time (often unknown)
- $\mathrm{C} \geq 0$: the **censoring time** (end of follow-up, dropout, etc.)

What we observe is:

$$Y = \min(T, C)$$

and an event indicator:

$$\delta = 1[T \leq C]$$

Interpretation:

- $\delta = 1$: the event is observed, so $\mathrm{T} = \mathrm{Y}$
- $\delta = 0$: the observation is censored, so we only know $\mathrm{T} > \mathrm{Y}$

With covariates x, the dataset is:

$$\{(Y_i, \delta_i, x_i)\}_{i=1}^{n}$$

where n is the number of individuals (sample size).

What we want to estimate

Depending on the task:

- **Descriptive (no covariates x)**
  Estimate the distribution of T, typically via:
  - Survival function
    $$S(t) = P(T > t)$$
  - Hazard function
    $$h(t)$$

- **Regression/prediction (with covariates x)**
  Model how the time-to-event depends on x, e.g., estimate:
  - Individualized survival curves
    $$S(t \mid x)$$
  - Relative risk effects (like in Cox models)

Why standard regression/classification fails

In many Machine Learning problems, the target is fully observed (a number or a label).
HOWEVER, in **Survival Analysis**, the true event time T is often **not observed** for everyone.
What we we observe is:

$$\mathrm{Y} = \min(T, C)\,, \delta = 1[T \leq C], \{(Y_i, \delta_i, x_i)\}_{i=1}^{n}$$

If $\delta = 0$, we only know $T > Y$ (partial information), not the exact time.

If you run standard regression on Y:

- you treat censored times as if they were true event times → **systematic bias** (typically underestimation of survival time)
- you ignore that some "labels" are incomplete

If you convert it into a classification task (event within a fixed horizon):

- you discard time information (two patients with events at week 2 and week 40 become the same label)
- results depend heavily on the chosen horizon

Takeaway: Survival Analysis needs methods that can learn from **partially observed time-to-event outcomes** (censoring-aware estimation and evaluation).

Core survival functions

- **CDF (Cumulative Distribution Function)/ Risk:** the probability the event has happened by time t (cumulative risk).

$$F(t) = P(T \leq t)$$

- **Survival function:** the probability of being event-free beyond t.

$$S(t) = P(T > t) = 1 - F(t)$$

Example: $S(12) = 0.8$ means "about 80% are expected to remain event-free after 12 units of time."

- **Density (continuous-time setting):** how probability mass is distributed over event times.

$$p(t) = \frac{d}{dt}F(t) = -\frac{d}{dt}S(t)$$

- **Hazard function:** the instantaneous event rate at time t, given survival up to t.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

Important: hazard is a rate, not a probability.

- **Cumulative hazard:** the total accumulated hazard (risk intensity) up to time t.

$$H(t) = \int_0^t h(u)\, du$$

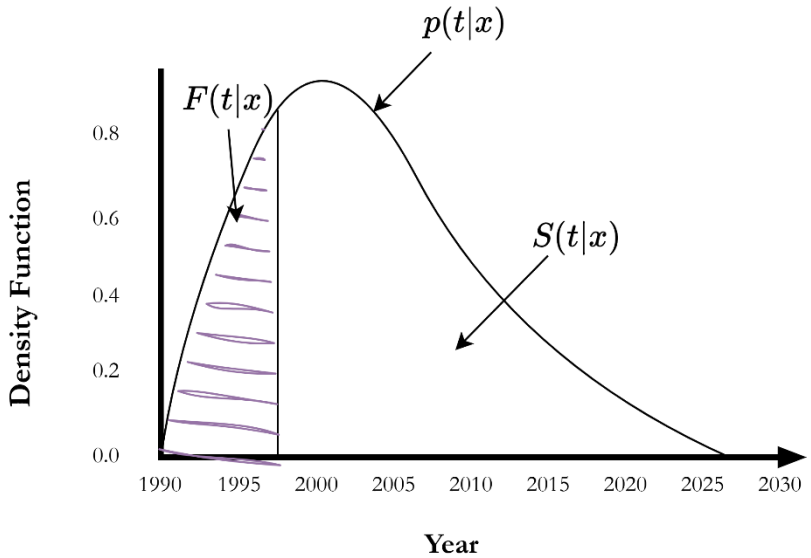
Core survival functions

Survival and hazard are just two equivalent ways of describing the same time-to-event distribution

$$S(t \mid x) = e^{-H(t|x)}$$

$$p(t \mid x) = h(t \mid x) S(t \mid x)$$

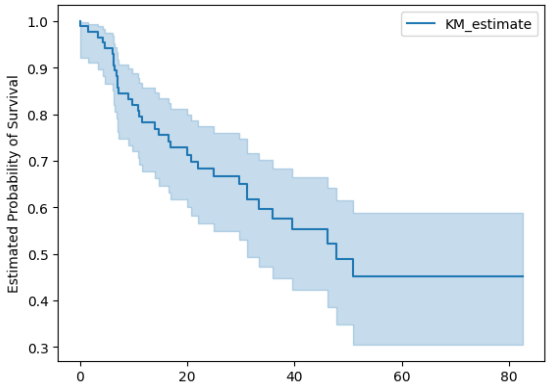
$$h(t \mid x) = -\frac{d}{dt} \log S(t \mid x)$$



Kaplan–Meier (KM) estimator

KM is a **non-parametric** estimator of the survival function $S(t)$.

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$



where:

- t_i: ordered event times
- d_i: number of events at time t_i
- n_i: number "at risk" just before t_i

Intuition

KM multiplies step-by-step conditional survival probabilities.
It produces a **step function** that drops at event times and stays flat between events.

Cox Proportional Hazards model (CoxPH)

$$h(t \mid x) = h_0(t)\exp(\beta^\top x)$$

- $h_0(t)$: baseline hazard (shared time pattern)
- $\exp(\beta^\top x)$: multiplicative effect of covariates

Hazard Ratio (HR)

For a one-unit increase in covariate x_j:

$$HR = \exp(\beta_j)$$

Interpretation:

- $HR > 1$: higher instantaneous risk → earlier events
- $HR < 1$: lower instantaneous risk → later events

Example: $HR = 1.2 \approx$ "20% higher hazard at any time," **if PH holds**.

Proportional Hazards (PH) assumption

The **ratio of hazards** between two individuals is **constant over time**:

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} = \exp(\beta^\top (x_a - x_b)) \quad \textit{(does not depend on t)}$$

Intuition

Risk can change over time for everyone (via $h_0(t)$), but **relative risk stays the same**.
If curves cross heavily or effects clearly change over time, PH may be violated.

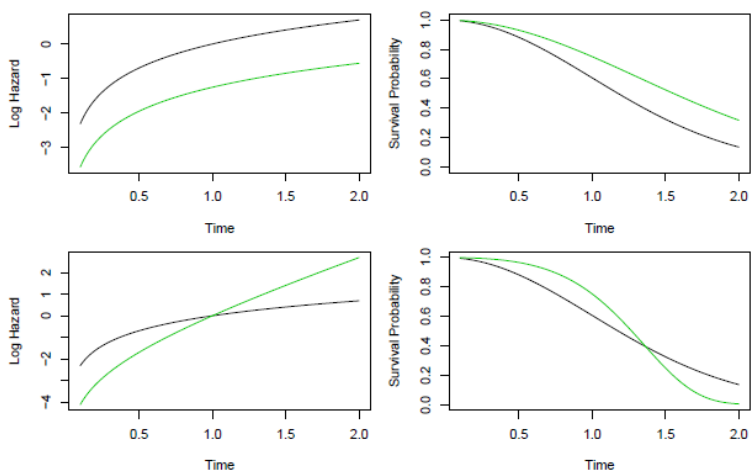


FIGURE 11.4. Top: *In a simple example with $p = 1$ and a binary covariate $x_i \in \{0, 1\}$, the log hazard and the survival function under the model (11.14) are shown (green for $x_i = 0$ and black for $x_i = 1$). Because of the proportional hazards assumption (11.14), the log hazard functions differ by a constant, and the survival functions do not cross.* Bottom: *Again we have a single binary covariate $x_i \in \{0, 1\}$. However, the proportional hazards assumption (11.14) does not hold. The log hazard functions cross, as do the survival functions.*